$$= 1(-40) + 4(-16) + -3(8) = -128 \quad \ldots (4)$$

From (3) and (4), we have

|A| obtained by expanding along R_1 is equal to expanding along C_1 .

Evaluation of determinant of order 3 by using Sarrus Rule (named after the French Mathematician Pierre Frédéic Sarrus)

Let $A = [a_{ij}]_{3\times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Write the entries of Matrix A as follows :

$$\begin{array}{ccccc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{array}$$

Then $|A|$ is computed as follows :

$$|A| = [a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}] - [a_{33}a_{21}a_{12} + a_{32}a_{23}a_{11} + a_{31}a_{22}a_{13}]$$

Example 7.16

Find $|A|$ if $A = \begin{bmatrix} 0 & \sin\alpha & \cos\alpha \\ \sin\alpha & 0 & \sin\beta \\ \cos\alpha & -\sin\beta & 0 \end{bmatrix}$.

Solution

$$\begin{vmatrix} 0 & \sin\alpha & \cos\alpha \\ \sin\alpha & 0 & \sin\beta \\ \cos\alpha & -\sin\beta & 0 \end{vmatrix} = 0M_{11} - \sin\alpha M_{12} + \cos\alpha M_{13}$$

$$= 0 - \sin\alpha(0 - \cos\alpha\sin\beta) + \cos\alpha(-\sin\alpha\sin\beta - 0) = 0.$$

Example 7.17

Compute $|A|$ using Sarrus rule if $A = \begin{bmatrix} 3 & 4 & 1 \\ 0 & -1 & 2 \\ 5 & -2 & 6 \end{bmatrix}$.

Solution

$$\begin{array}{ccccc} 3 & 4 & 1 & 3 & 4 \\ 0 & -1 & 2 & 0 & -1 \\ 5 & -2 & 6 & 5 & -2 \end{array}$$

$$|A| = [3(-1)(6) + 4(2)(5) + 1(0)(-2)] - [5(-1)(1) + (-2)(2)3 + 6(0)(4)]$$

$$= [-18 + 40 + 0] - [-5 - 12 + 0] = 22+17 = 39.$$

Note 7.6

For easier calculations, we expand the determinant along a row or column which contains maximum number of zeros.

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